

AI Aptitude Assessment

Problem Statement & Use Cases

Digital Learning Transformation Initiative | 2026

- Problem Statement 1
- Overarching Positioning..... 2
- Learner Use Case Scenarios 2
 - 1. Behavioral Readiness Assessment (Education Sector)..... 3
 - 2. Technical Competency Verification (Commercial & Public Sector) 3
 - 3. Adaptive Pre-Assessment & Personalized Learning Entry 3
- Tech / Experience Considerations..... 4
- Value Proposition..... 5
- Right to Win (Strengthened & Differentiated)..... 5
 - 1. Proprietary Content Foundation 5
 - 2. Instructional Design Expertise..... 5
 - 3. Sector-Specific Contextual Depth (25+ Years)..... 5
 - 4. Rigorous, Auditable Assessment Architecture..... 6
 - 5. Closed-Loop Learning Integration..... 6
- Strategic Positioning (Clear Contrast with Generic Alternatives) 6
- Foundation for Future Transformation..... 7

Problem Statement

Organizations invest in high-quality, research-backed training to drive behavior change across education, commercial, and public sector workforces. Yet the primary mechanism for measuring that impact remains a multiple-choice knowledge check — a post-course exam that tests whether a learner can recall information, not whether they can apply it.

This gap between knowing and doing is well understood in learning science, but it has no structural solution in today's digital training landscape. Pass/fail knowledge exams are easy to complete on autopilot, increasingly vulnerable to AI-assisted completion, and structurally incapable of revealing the skills, beliefs, and behavioral dispositions that training is actually designed to develop.

The result is a measurement problem with real consequences: learners earn completion credits without demonstrating competence, administrators see pass rates instead of readiness, and organizations have no defensible picture of whether their training investment is producing the workforce capability or institutional culture they need.

Despite holding deep content expertise, validated learning objectives, proprietary assessment item banks, and 25+ years of trusted sector relationships, Vector is limited in how we **evaluate whether learners have actually internalized the skills, beliefs, and competencies** our training was designed to develop.

Overarching Positioning

Enable organizations, agencies, and institutions to move beyond knowledge verification toward genuine competency measurement — using scenario-based, conversational assessments grounded in course content, validated learning objectives, and sector-specific contexts that reveal what learners can actually *do*, not just what they can recall.

This is not a replacement for traditional assessment — it is an **extension of the learning experience into a richer, more defensible, and more actionable measure of learner readiness and aptitude**.

Learner Use Case Scenarios

1. Behavioral Readiness Assessment (Education Sector)

- An administrator completes a sexual harassment prevention course and faces a scenario: “A colleague describes a situation that sounds like it may have involved non-consensual contact. What do you do?”
- A high school student finishing a substance misuse course is prompted: “You’re at a party and someone is pressuring a friend to drink. Walk me through how you’d respond.”
- A student completing a mental health awareness course engages with a scenario involving a peer showing signs of crisis and is assessed on recognition, empathy, and appropriate action.

 Assessment reveals:

- Whether the learner can identify a problematic behavior — not just define it
- Whether they have internalized appropriate response strategies
- Where they are strong and where they need reinforcement — surfaced back to them as a personal competency profile

2. Technical Competency Verification (Commercial & Public Sector)

- An industrial worker completing a hazardous materials handling course is walked through a scenario involving an unexpected spill and asked to describe the steps they would take — assessed against a rubric tied to OSHA standards.
- A construction worker finishing a workplace safety course encounters: “You arrive on site and notice a colleague is not wearing appropriate PPE near an active electrical panel. What do you do, and in what order?”
- A firefighter completing an NFPA-accredited training is assessed on their ability to apply protocol in a multi-variable incident scenario before being issued a micro-credential.

 Assessment delivers:

- A construct-mapped score tied to specific skills covered in the course (e.g., 10 competency constructs, scored individually)
- A defensible, auditable record of the learner’s responses and how they map to required competencies
- An aggregate view for administrators showing team-level readiness across safety-critical skills

3. Adaptive Pre-Assessment & Personalized Learning Entry

- A learner begins an accredited first responder course and is assessed upfront on their baseline competency — before engaging with any content.
- Based on their responses, the system:
 - Directs them to pay closer attention to constructs where they showed gaps (and the personal benefits of making improvements in these areas)
 - Tests them out of content areas where they demonstrated strong existing knowledge
 - Elevates the sophistication of content in areas where they are already competent
- At course completion, a post-assessment generates a before/after competency profile that demonstrates growth and identifies remaining development areas.

👉 **This creates:**

- A more efficient, personalized learning experience that respects what learners already know, provides tailored feedback, and builds intrinsic motivation for training participation
- A stronger evidence base for learning impact — tied to measurable competency growth, not just completion
- A foundation for competency-based credentialing tied to demonstrated skills, not just seat time

Tech / Experience Considerations

- Seat time must be actively managed — the conversational assessment should replace or reduce other content or assessment modalities rather than add to total training time
 - Consider voice/narrated response capability to reduce typing burden
 - Consider retiring or shortening traditional post-course exam items where the conversational assessment covers the same constructs
- The assessment must coexist with prescribed exam requirements — accredited trainings from licensing bodies may require specific exam questions to remain intact; the aptitude layer sits alongside, not in place of, mandated items
- Rubric-based scoring mapped to specific constructs (e.g., 10 constructs assessed per conversation, each scored individually) is critical for customer-facing defensibility around the rigor and output of the assessment
 - We are considering both quantitative and qualitative scoring approaches for Aptitude Assessment. Here's an example of qualitative scoring:
 - Below Average / Practice Needed: you could use some improvement and we'll generate/adapt content to get you to an average/good state.
 - Average / Good: You could start some practice to go beyond this level, but you don't need to. This is a "good" level to be at.
 - Above average / Excellent: You've gone above and beyond and don't need to do more practice. Some of your upcoming lessons have been tightened up since you've already mastered topics they include.
 - Aptitude Assessment would ultimately need to capture a lot of different constructs for it to serve as a pass/fail function or to trigger testing out or testing up around certain concepts... In that case, if the aptitude assessment scenarios and prompts cover a breadth of course-related learning objectives, then each of them could be scored with a below/average/above as you're proposing but then you need to get at least

- average on 80% of the objectives to "pass" -- it gives us both qualitative and quantitative
- Learner-facing competency profile is key for creating engagement and intrinsic/extrinsic motivation: areas of strength, areas for growth, and personalized developmental feedback
- Administrator dashboard: must be able to show aggregate competency data across learner groups, individual profiles, and construct-level score mapping
- Full auditability: administrators can view the complete chain of prompts and responses and trace scores back to specific constructs
- Admin preview and approval: administrators should be able to review and approve the conversation approach before deploying it to their learners
- AI-completion resistance: the open-ended, conversational format is inherently harder to game than multiple-choice exams where answer keys can be found online

Future Considerations

- Micro-credentialing tied to demonstrated competency thresholds
- Personalized learning pathways driven by pre-assessment data
- Organizational competency benchmarking across roles, departments, and time
- Integration with the AI Knowledge Assistant to close the loop between real-time performance support and competency measurement

Value Proposition

Move beyond the pass/fail knowledge check to deliver a rigorous, contextually-grounded picture of learner competence — one that produces actionable, individually meaningful insight for learners and strategically relevant data for organizations about where capability is strong and where investment is needed.

Right to Win (Strengthened & Differentiated)

The advantage is not just the ability to build a conversational assessment engine — it is the depth of validated content, instructional expertise, and organizational trust that makes that engine rigorous, defensible, and genuinely sector-relevant.

1. Proprietary Content Foundation

- Ownership of the underlying course scripts, learning objectives, assessment item banks, answer keys, and feedback responses for incorrect answers across all product lines
- This provides the raw material for building scenario prompts and rubrics that are grounded in validated instructional content — not synthesized from the open internet

2. Instructional Design Expertise

- In-house instructional designers with sector-specific knowledge who can create rigorous, contextually appropriate scenario prompts and competency rubrics
- Ability to map assessments to specific constructs across multiple domains: knowledge, beliefs, skills, and behavioral intentions
- Competency frameworks built with the same rigor applied to course and assessment design

3. Sector-Specific Contextual Depth (25+ Years)

- Longstanding partnerships across K-12, higher education, commercial (AEC, industrial), and public sector (first responder, government) organizations
- Deep understanding of the regulatory, accreditation, and compliance contexts that shape what “defensible assessment” means in each sector
- Institutional trust that a generic assessment tool cannot replicate

4. Rigorous, Auditable Assessment Architecture

- Construct-mapped scoring: every response traces back to a specific learning objective or competency area covered in the course
- Full chain-of-evidence auditability: administrators can see the complete prompt-response record and verify how scores were derived
- Assessments designed to be defensible for accreditation, licensing, credentialing, and compliance-driven use cases

5. Closed-Loop Learning Integration

- Pre-assessment → personalized learning → post-assessment → competency profile → organizational insight

👉 ***This creates a system where:***

- Learner profiles reflect genuine competency growth, not just completion records and knowledge gains
- Administrators see strategically relevant data tied to business-critical capability gaps
- Assessment data feeds back into content improvement and learning program design

Strategic Positioning (Clear Contrast with Generic Alternatives)

Today’s Multiple-Choice Assessment:

- Tests knowledge recall, not skill application
- Easy to game — answer keys are findable online; AI agents can complete exams automatically
- Produces pass/fail outputs with no individual or organizational insight

- Structurally disconnected from the behaviors training is designed to build

A Generic AI Assessment Tool:

- Can generate scenario-based questions but lacks grounding in validated course content
- No connection to specific learning objectives, organizational context, or sector requirements
- No rubric-based construct mapping; no auditability chain
- Not defensible for accreditation, licensing, or compliance use cases

The AI Aptitude Assessment:

Grounded in proprietary course content, instructionally designed for sector-specific competency, mapped to measurable constructs, fully auditable, and built to deliver individually meaningful and organizationally strategic insight — not just a completion record.

Foundation for Future Transformation

This initiative is not just a product feature — it lays groundwork for:

- Competency-based progression models tied to demonstrated skills, not time-in-seat and right/wrong answers
- Personalized learning pathways driven by pre-assessment competency data
- Micro-credentialing frameworks that recognize and validate specific skill sets
- Organizational competency benchmarking across roles, departments, and cohorts over time
- AI-enhanced adaptive content delivery that responds dynamically to learner capability
- Integration with the AI Knowledge Assistant to create a continuous performance and competency loop

 **In other words:**

We move from “training completed” → “competency demonstrated.”